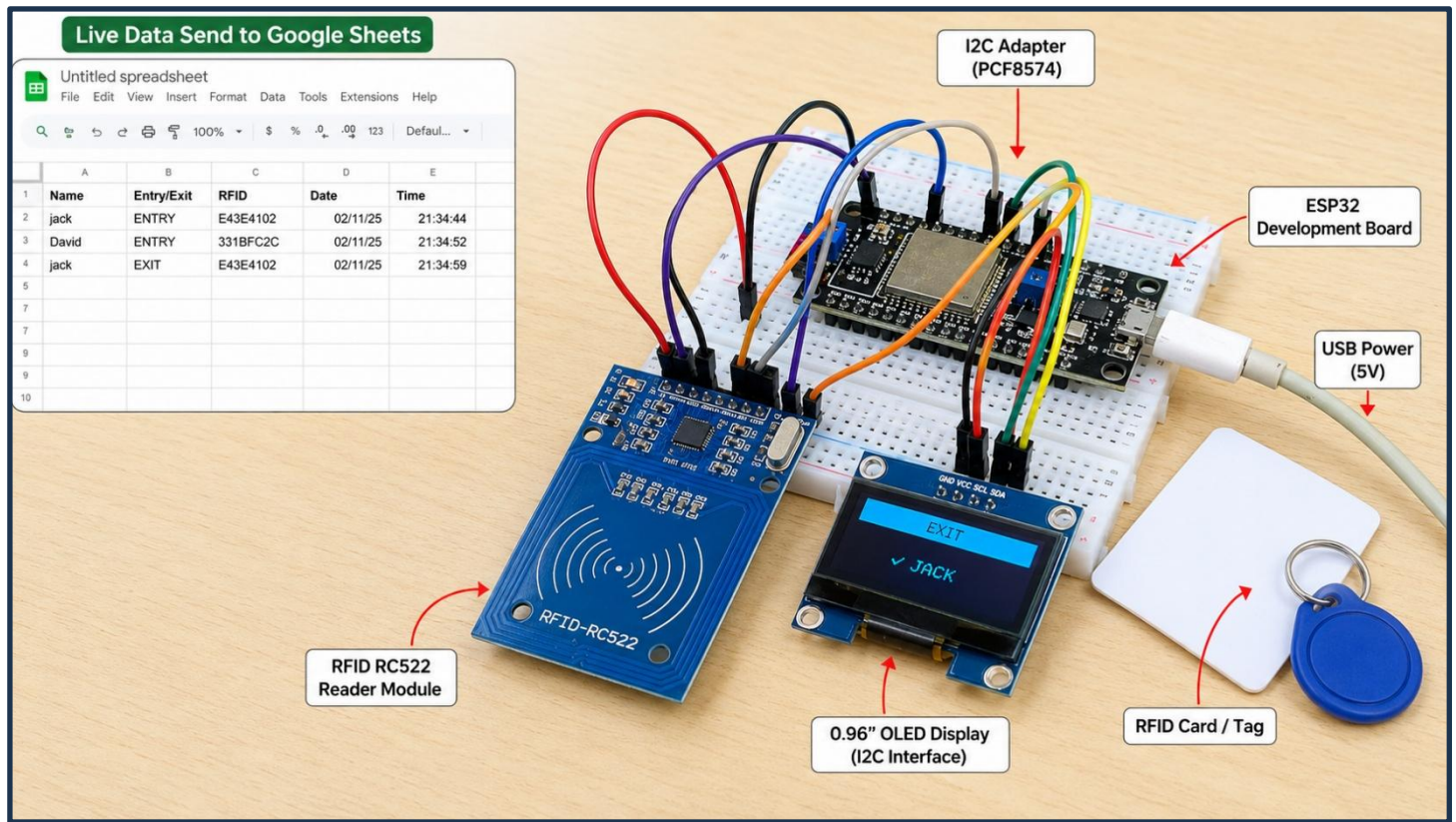
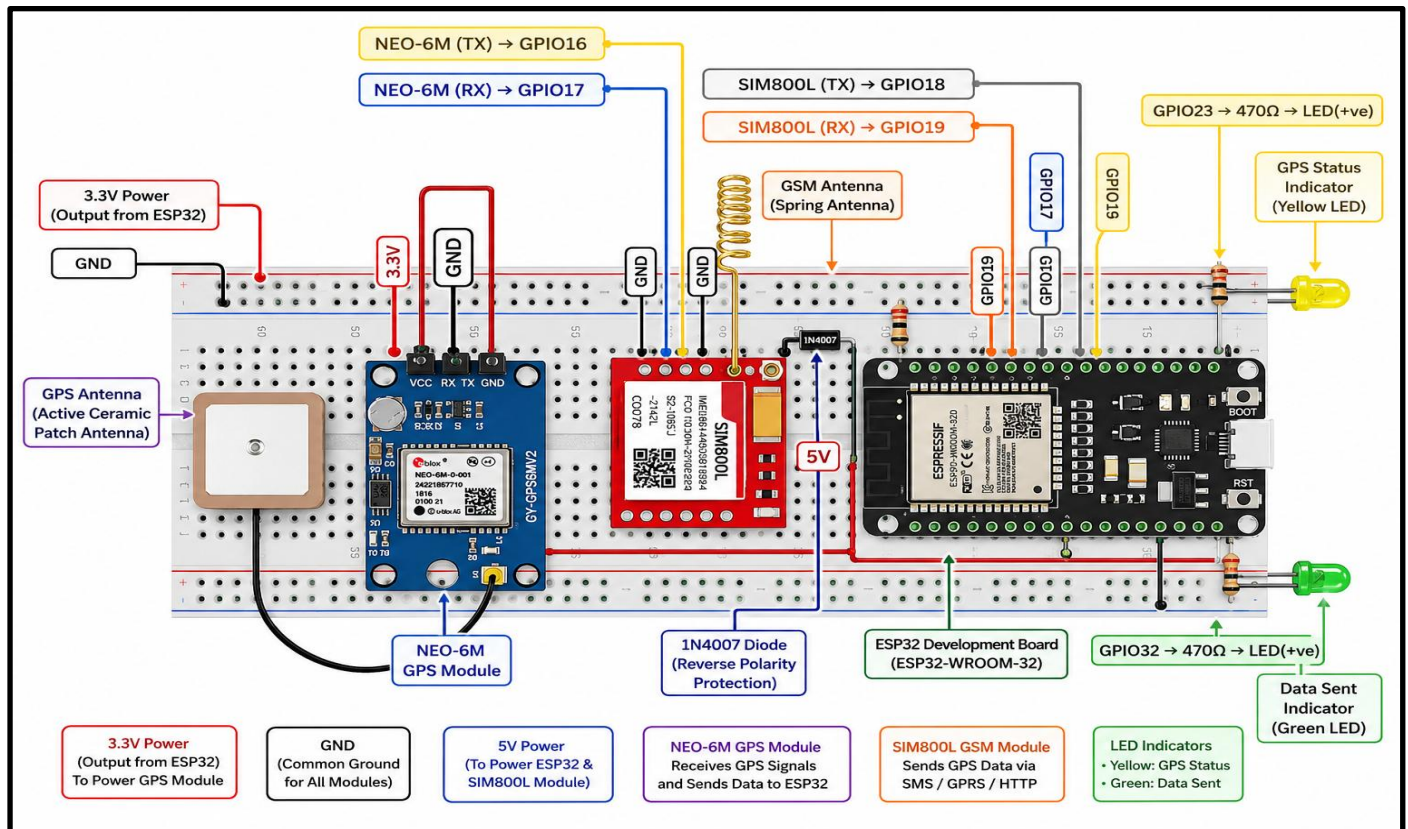


RFID BASED ATTENDANCE SYSTEM WITH GOOGLE SHEET



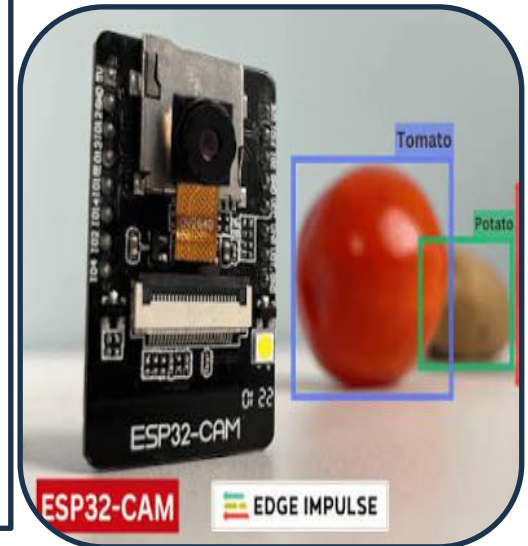
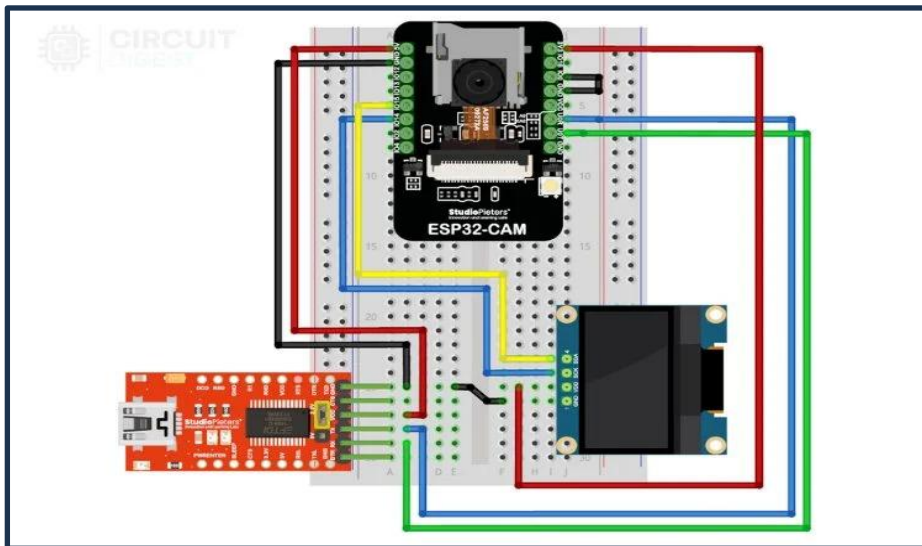
COMPONENTS	SPECIFICATION	COST	PURPOSE
ESP32 Development Board	ESP32-WROOM-32, Dual Core, Wi-Fi + Bluetooth, 240 MHz		Main controller that reads RFID data, updates OLED, and uploads attendance to Google Sheets through Wi-Fi.
RFID RC522 Reader Module	MFRC522, 13.56 MHz SPI Interface		Reads RFID cards/tags and sends UID to ESP32.
RFID Card / Key Tag	MIFARE Classic 1K, 13.56 MHz		Used as the attendance card. Each card has a unique UID.
0.96-inch OLED Display	SSD1306, 128×64 Pixels, I2C Interface		Displays student name, ID, Entry/Exit status, Wi-Fi status, etc.
Breadboard	830 Tie Points (or Mini Breadboard)		Used for making temporary connections without soldering.
Jumper Wires	Male-to-Male Dupont Wires		Connects ESP32, RFID reader, OLED, and power.
USB Cable	USB Type-C or Micro USB (depends on ESP32 board)		Powers the ESP32 and uploads the program

GPS TRACKING SYSTEM USING ESP32



COMPONENTS	SPECIFICATION	COST	PURPOSE
ESP32 Development Board	ESP32-WROOM-32		Main microcontroller that processes GPS data and sends it through GSM.
NEO-6M GPS Module	u-blox NEO-6M		Receives satellite signals and provides latitude, longitude, speed, date, and time.
GPS Ceramic Antenna	Active GPS Patch Antenna		Connected to the NEO-6M module to receive GPS satellite signals.
SIM800L GSM/GPRS Module	Quad-band GSM Module		Sends GPS location data via SMS, GPRS, or HTTP over the cellular network.
GSM Spring Antenna	GSM/GPRS Antenna		Improves GSM network signal reception for the SIM800L module.
Breadboard	Solderless Breadboard		
Jumper Wires	Male-to-Male Dupont Wires		Used for all module interconnections.
470 Ω Resistor (2 Nos.)	1/4 Watt		Current limiting resistor for LEDs.
1N4007 Diode	Rectifier Diode		Protects the SIM800L power circuit from reverse polarity or voltage spikes.

OBJECT DETECTION USING ESP32-CAM



<u>components</u>	<u>specifications</u>	<u>COST</u>	<u>purpose</u>
ESP32-CAM Development Board	AI Thinker ESP32-CAM (ESP32-S)		Main controller with built-in Wi-Fi, Bluetooth, and OV2640 camera for image capture and streaming.
OV2640 Camera Module	2 MP Camera		Captures images and video for object detection or monitoring. (<i>Pre-installed on the ESP32-CAM.</i>)
0.96-inch OLED Display	SSD1306, 128×64 pixels, I2C		Displays device status, IP address, detection results, or system information.
FTDI USB-to-Serial Programmer	USB to TTL (3.3V/5V selectable)		Used to upload programs from a computer to the ESP32-CAM.